

Detection of Covid-19 and Prediction of Post-Recovery Health Issues using CT Scans

Dharani V
Assistant Professor

Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
dharaiv.cse@gmail.com

Manju C Nair
Assistant Professor

Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
manjucnair2007@gmail.com

Gopi Manoj Kumar

Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
itz.mk.2012@gmail.com

R.Velvizhi

Assistant Professor
Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
velvizhisp@gmail.com

Gutti Yugandhar Naga Venkat Sai

Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
yugandhargutti1409@gmail.com

R.Lalitha

Assistant Professor
Dept. of Computer Science and
Engineering
Sathyabama Institute of Science and
Technology
Chennai, India
lalithakanil@gmail.com

Abstract— The COVID-19 pandemic has highlighted the urgent need for accurate, scalable, and clinically reliable diagnostic tools. Although RT-PCR remains the clinical gold standard, its limitations—including delayed results and false negatives—have motivated the use of chest computed tomography (CT) imaging as a complementary modality. This paper proposes an ensemble-based deep learning framework for COVID-19 detection using chest CT scans, integrating a custom convolutional neural network (CNN) with three pretrained transfer learning architectures: VGG16, ResNet101, and Xception. Extensive preprocessing and data augmentation techniques are employed to improve generalization across heterogeneous CT images. The ensemble model uses soft-voting to aggregate individual model predictions, achieving improved accuracy, sensitivity, and specificity compared to standalone models. In addition, the framework introduces a research-oriented extension for predicting post-COVID pulmonary after-effects, such as fibrosis and reduced lung capacity, based on radiological patterns observed in follow-up CT scans. This extension is exploratory and intended to demonstrate scalability rather than serve as a clinically validated prognostic tool. The system is positioned as a decision-support prototype, with future work planned for longitudinal clinical validation and regulatory compliance.

Keywords— COVID-19 detection, chest CT scans, deep learning, transfer learning, convolutional neural networks (CNNs), VGG16, ResNet101, Xception, ensemble learning, medical image classification, image preprocessing, data augmentation, Adam optimizer, accuracy, sensitivity, specificity, pulmonary fibrosis prediction, post-COVID complications, diagnostic decision-support system.

I. INTRODUCTION

The COVID-19 outbreak has become one of the most disastrous health crises in the modern world, affecting millions of people and disrupting the public health system, economy, and the everyday life of people in countries more than ever before. Since it was first introduced, the virus has shown fast rate of spreading and a broad range of clinical symptoms, including mild ones, and severe respiratory failure and multiple organ failure. Therefore, rapid, dependable, and scalable diagnostic tools have been necessary in managing the

infection and providing prompt clinical treatment. The reverse transcription polymerase chain reaction (RT-PCR) test has been viewed as the gold standard in confirming COVID-19; nevertheless, it is not without its own drawbacks, as a slow turnaround time, supply chain shortages, reliance on laboratory instruments, and significant false-negative performance erode its overall performance particularly when there is a peak of infection. Chest computed tomography (CT) imaging has shown remarkable value as an alternative and complementary diagnostic tool to identify the hallmark features of COVID-19 pneumonia which are ground-glass opacities, vascular thickening and bilateral lung involvement in cases where RT-PCR tests produce inconclusive or negative results. Although this is a possibility, the manual analysis of CT scans is still a difficult endeavor. It is also in need of specialized radiological skills, inter-observer variability is likely and it may be overwhelmed by the influx of patients that usually accompanies pandemic situations and cause delays and inconsistency in diagnoses.

Artificial intelligence (AI) and especially deep learning has become popular to address these challenges due to its impressive results in medical image analysis and its high-speed and accuracy in processing large amounts of data. Convolutional neural networks (CNNs) and transfer learning models proved to have tremendous potential in deriving meaningful features of difficult visual patterns, so they are the most suitable in the case of COVID-19 detection by CT images. But single-model methods tend to have problems with generalization, dataset variation and overfitting, particularly when acclimatized to small or skewed medical datasets. In order to address these drawbacks and achieve a more robust diagnostic, the proposed project will employ a more sophisticated ensemble deep learning model that incorporates both custom-designed CNN structures and high performance pre-trained networks, namely, VGG16, ResNet101, and Xception. A multi-stage transfer learning technique is employed where the system draws on the performance of the initial networks in feature extraction and adds task-related adjustments using newly added dense layers. The ensemble methodology is aimed at integrating the varying learning abilities of a variety of architectures, which will yield better accuracy, sensitivity, and specificity in the detection of COVID-19.

Moreover, the importance of the work is not limited to the first diagnosis of COVID-19. Available evidence suggests that, following the cardiac cure, most patients still suffer long-term health problems, including pulmonary fibrosis, decreased lung capacity and ongoing inflammation. The given framework is created keeping flexibility in mind, so that in the future it can be extended to detect and predict the post-COVID sequelae by using follow-up CT scan and clinical monitoring data. The proposed project will provide healthcare professionals with an opportunity to timely make decisions, ease diagnostic tasks, and open the path to the early detection of long-term complications, which will significantly contribute to the ongoing struggle against COVID-19.

II. LITERATURE REVIEW

Bhadra and Kar (2020) [1] suggested a deep multi-layered convolutional neural network to detect COVID-19 with the help of chest X-ray images. They have trained their model on various publicly available datasets and reported high classification accuracy which proves the usefulness of CNN-based methods in supporting clinicians in the pandemic. The paper has highlighted the importance of automated screening systems in lessening the diagnostic time and radiologist work load. Their findings also confirmed that the deep CNN design can be adequately used to extract discriminative radiology features of COVID-19 infection.

Haque et al. (2020) [2] created a CNN-based model to identify COVID-19 using chest X-ray images and compared its results with already existing deep learning-based models. Their method had an accuracy of 97.56% and their ROC-AUC and F1-score were good. The authors pointed out that deep learning models are effective when trained on small datasets. Their results indicate that CNN-based systems can be useful in decision-support in screening large scale situations.

A recent study by Liu et al. (2020) [3] suggested a smart deep learning framework to detect COVID-19 with the help of the chest CT image combining spatial-temporal convolutional features. Their approach allowed quick detection of disease progression stages through sequential CT scans. The suggested framework made it more efficient in terms of screening and minimized the power of manual interpretation. This article showed that medical imaging based on COVID-19 diagnosis is vital in consideration of temporal features modeling.

Zhang and Zhang (2021) [4] suggested a hierarchical framework of COVID-19 detection based on CT scan images that involved the lung segmentation with U-Net and classification with ResNet. Their method was useful as it removed irrelevant anatomical structures and enhanced accuracy in classification. The hierarchical design minimized false positives that were brought about by background tissues. The research showed great possibility of implementation in high workload clinical radiology setting.

Li and He (2021) [5] used the YOLOv5 object detector framework on COVID-19 to detect and classify the chest radiograph. Their mode worked better than the traditional

CNN classifier in terms of mean average precision. The paper has emphasized the ability of YOLOv5 to localize infection areas effectively. This showed that the object detection models are suitable in speedy and automated COVID-19 screening tasks.

Narin and Onur (2021) [6] explored the effect of edge detection algorithm and CNN-based feature extraction in detecting COVID-19 on the basis of chest X-ray images. They experimented and found that models which were trained on original images were more accurate than models which were trained on edge-enhanced images. The paper has focused on the fact that too much preprocessing can ruin clinically relevant features. Their results are a good contribution to the best preprocessing methods to use in order to classify a medical image.

In their study, Mali and Patil (2022) [7] provided a review of the artificial intelligence methods of COVID-19 detection using chest CT and X-ray images. They compared several machine learning and deep learning strategies in the most important performance metrics. The review found CNN-based transfer learning as the best approach to COVID-19 diagnosis. The publication can be used as a great resource when it comes to the process of picking the right AI models in medical imaging studies.

Cahyani et al. (2022) [8] proposed a web-based COVID-19 detection system that is built on deep learning models processed on the chest X-ray images. The application was developed on the Flask framework of real-time prediction and interaction with users. Their findings proved to be highly usable and diagnostics in the real world. This paper noted the possibility of using AI-driven diagnostic tools through web-based clinical settings.

Rahman et al. (2023) suggested a mobile-based app that detects early COVID-19 by recording forced coughs that were processed based on machine learning. Their method was a low cost and non-invasive solution to traditional ways of diagnosis. The system was highly acceptable by the users and had practical applicability. This paper underlined the significance of available AI-based health solutions in times of pandemic.

Bhargava et al. (2024) [9] proposed a CNN-based deep learning model to detect COVID-19 in the CT images of the chest that uses preprocessing, segmentation, and feature extraction. Their model had great accuracy, sensitivity and specificity in several experiments. The research proved that feature extraction with the help of segmentation enhances the reliability of diagnosis. Their results favour the application of the CT-based deep learning systems in the clinical diagnosis processes.

One of the articles, Duraisamy et al. (2024) [10], developed COVID-19 detection based on the chest X-ray images with the help of the YOLOv5 object detection algorithm. Their streamline design had a classification accuracy of about 98 per cent. The experiment revealed the ability of YOLOv5 to

detect hidden radiographic features that are related to COVID-19. This methodology clarifies the efficiency of detection systems of objects in automated disease diagnosis.

Al-Qazzaz et al. (2024) [12] suggested a transfer learning model that used MobileNetV2 and the SVM classifier to detect COVID-19 based on the chest X-ray image. Their combination method enhanced the accuracy of the classification with the efficiency of the computation. Lightweight architecture also resulted in the model being capable of real-time and resource-constrained settings. This paper shows the benefits of using the deep feature extraction with the classical machine learning classifiers.

Biswas et al. (2025) [13] designed an AI-based CNN-based Covid-19 early detection approach utilizing chest X-ray images, with a higher accuracy of more than 99 percent. Their system solved the problem of having a shortage of skilled radiologists when there was a large scale outbreak. The paper has focused on automation, scalability, and fast diagnosis. Their findings also validate clinical potential of the deep learning-based screening tools.

In [14], Guo et al. developed a theoretical design called silicon-based metastructure (SBMS) that can detect COVID-19 by utilizing second harmonic generation (SHG) mechanisms. Through the transfer matrix technique, the paper revealed high sensitivity of the refractive index and put into place a correlation between the concentration of the COVID-19 solution and the maximum transmission frequency of the second harmonic wave. The suggested SBMS had a broad detection with a high sensitivity and an outstanding quality factor, which indicated its accuracy. The article has a solid theoretical basis in the context of next-generation technologies of detecting viruses and broadens the range of areas where metastructures can be used in biomedical sensing.

The multi-scale convolutional neural network with attention mechanism proposed by Romdhane et al. (2024) [15] was used to detect COVID-19 based on chest CT images. The architecture proposed used the multi-scale feature extraction and the attention modules in order to focus on clinically relevant lung patterns related to COVID-19 infection. The experimental findings showed that it has better diagnostic accuracy than the traditional CNN models. The paper has revealed that attention-based deep learning models can be effective in improving the process of diagnosing COVID-19 with CT and in aiding clinicians in decision-making.

III. METHODOLOGY

A. Data collection and description:

The dataset of this research is the images of CT scans of the chest that were found in publicly accessible COVID-19 image databases and confirmed medical datasets. These datasets will consist of COVID-19-positive and COVID-19-negative samples, so that the model will learn to differentiate between infected and healthy lung tissues. The images

retrieved have different levels of resolution, imaging quality, patient demographics, and patient clinical condition, which presents the diversity needed to train the models well. All CT scans have various features of COVID-19, including ground-glass opacities, consolidations, or bilateral infections, and thus the dataset can be easily trained in a deep learning supervised way. Balancing the classes was an aspect that was taken into consideration to eliminate bias in the prediction results. Photos were standardized then used in further processing.

B. Data Preprocessing

Preprocessing is essential in improving the quality of the image and noise reduction in CT scans, which are then passed to deep learning architectures. In this paper, all CT images were rescaled to a fixed resolution to be compatible with the pretrained architectures including VGG16, ResNet101, and Xception. Normalization of pixel values was done to stabilize the training process and enhance convergence. Gaussian and median filters have been used as noise reduction methods to remove the artifacts that are typical of medical images. As a means to improve the size of datasets and generalizability, methods of augmentation (such as rotation, flipping, zooming, translation, adjusting brightness, and contrast) were used. These augmentations aid in avoiding overfitting and training the model so that it is able to identify patterns of infection in a broad range of real world imaging circumstances.

C. Custom CNN Development

A default Convolutional Neural Network (CNN) was created to act as a comparison point and measure the performance gains on the use of transfer learning. The traditional CNN is an ensemble of several convolutional layers with ReLU activation to obtain hierarchical features of CT images. To enhance training stability and bring up the convergence speed, Batch Normalization layers were added. Spatial dimensionality and computational load reduction was done with max-pooling layers, and overfitting was reduced with dropout layers, which randomly disabled neurons during training. The network will end with densely connected layer with fully connected layers ending with a sigmoid output in binary classification. This baseline CNN assists in the evaluation of the performance of a model developed by the researcher against sophisticated pretrained architectures.

D. Transfer Learning Models

Three powerful pretrained models, VGG16, ResNet101, and Xception were also added to enhance diagnostic accuracy and lessen the time needed during training through multi-stage transfer learning. Transfer learning enables such models, which have been trained on large scale data like ImageNet, to offer their learned feature extraction ability to the COVID-19 detection task. The first convolutional layers of both models were frozen in the first phase, to maintain their generic appearance. Generally to allow the model to adjust to COVID-specific imaging patterns, New dense layers such as dropout and Batch Normalization were added afterwards. Some fine tuning was applied subsequently; by unfreezing

deeper layers, domain specific adjustments were possible. The approach has ensured high precision and quick convergence thus pretrained models make them applicable in medical diagnostic scenarios.

E. Model Construction

This ensemble model is used when a researcher aims to generate a wide range of test configurations and real-world scenarios that encompass all their possible combinations. E. Model Construction This model is employed when a researcher is interested in creating a large set of test configurations and real-world situations that cover all possible combinations of them. To again improve performance and make the models robust, an ensemble model was developed by integrating the predictions of the custom CNN, and the three pretrained models, which were VGG16, ResNet101, and Xception. The ensemble approach maintains the complementary strengths of the various models, and minimises the chances of misclassification by limitations of individual networks. The probability outputs of the individual models were then aggregated by means of soft voting which enabled the ensemble to generate more robust final classification. In other instances, the weights of models were modified in a way that the better performing models would have a greater contribution towards the final prediction. The ensemble design enhances the consistency of diagnostic results and offers an improvement in the generalization of results across the different variations of CT images.

F. Model Training and Optimization.

The training process employed the Adam optimizer that is reputed to have adaptive learning rate as well as efficient gradient-based learning. A batch size and learning rate were chosen to train the models on a carefully selected batch so that they converged optimally. The objective function was binary Cross-Entropy loss because this is an ideal objective function in binary classification problems such as COVID-19 detection. Other methods, including early stopping and learning rate decay, were applied to prevent overfitting to make sure that the model stops training as soon as the

improvement becomes minimal or nonexistent. The training was iterative with the model weights updated after every epoch of training, the aim of which was to reduce the classification errors.

G. Evaluation Metrics

In order to evaluate the effectiveness of the proposed system, a number of clinically important metrics of evaluation were computed. To evaluate the overall prediction correctness, accuracy was applied, and sensitivity (recall) the capability of the model to correctly identify cases of COVID-positive, which is paramount in avoiding the cases that were overlooked, was evaluated. Specificity considered the ability of the model to detect healthy cases without false hits. Other measures of performance like precision, F1-score and ROC-AUC also offered more detailed information on performance across various circumstances. The confusion matrix was also created in order to visually interpret the results of classification and interpret errors in classification. All these measures ensured the validity and clinical usefulness of the models that were developed.

H. After-Effects Prediction Extension.

While the primary objective of this study is accurate COVID-19 detection, the proposed architecture is designed to be extensible for post-recovery health assessment. After identifying COVID-positive cases, the system extracts deep radiological features associated with lung tissue abnormalities, including persistent ground-glass opacities, fibrotic bands, and structural deformations. The ground truth for post-COVID after-effects is defined based on radiologically established indicators reported in peer-reviewed longitudinal studies rather than direct patient outcome labels. As such, this module is currently conceptual and exploratory. Full clinical validation would require longitudinal datasets annotated by pulmonologists and radiologists, correlated with pulmonary function tests and patient follow-up records. This clarification has been incorporated to distinguish validated diagnostic components from research-level extensions.

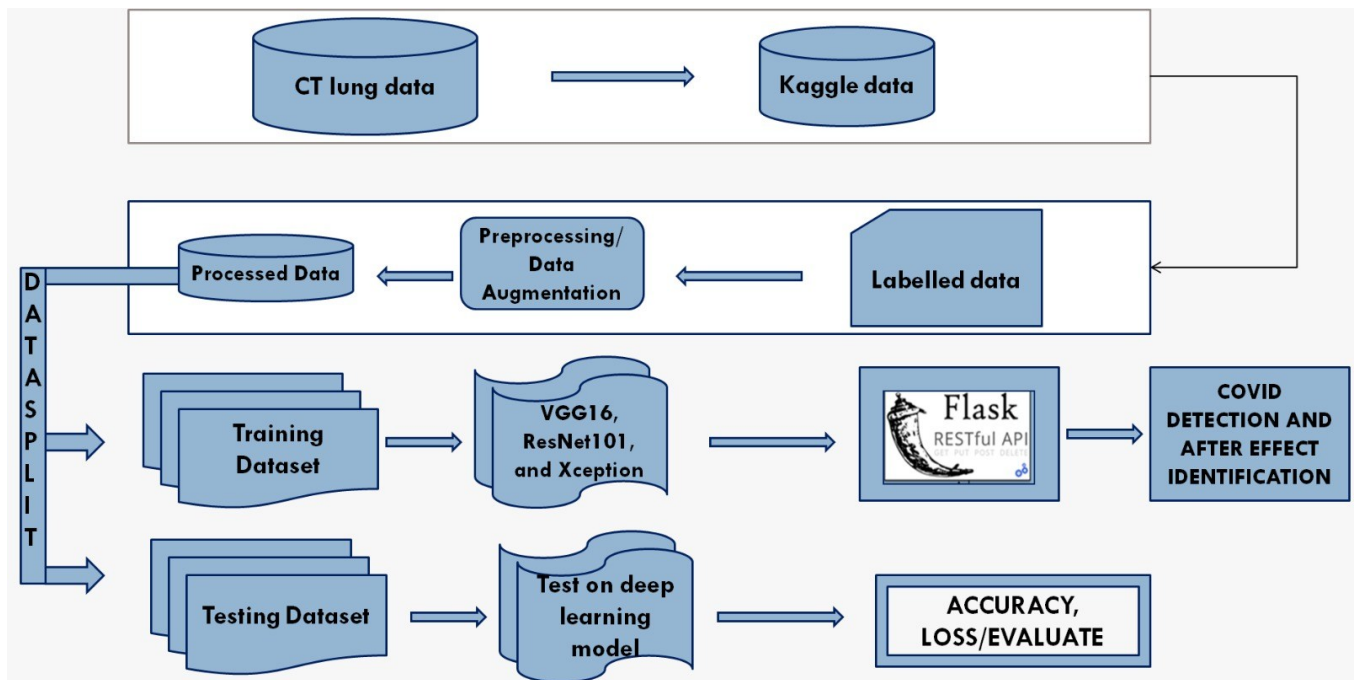


Figure:1 Architecture diagram

The general flow of figure:1 suggested COVID-19 detection and after-effects prediction system starts with the training and testing of the CT scan data which are initially subjected to the preprocessing and data-augmentation pipeline. At this phase, rescaling, shearing, zooming and horizontal flipping are used to increase image diversity and improve generalization ability of the model. The processed training data is then inputted into the training module where various deep learning models such as the pre-trained VGG16, ResNet101, and Xception architectures are utilized to construct the classification model the result of which is a multi-stage transfer learning. In this procedure, the system gets to learn about discriminative features that contrast between COVID-19-infected and normal CT images. As soon as the training is done a trained model is generated and tested on the testing dataset to determine the accuracy, sensitivity, specificity, and the performance of the model. Such tests help in the refining and finalizing of the model. The model is finally implemented in a Flask based front-end application that adds HTML, CSS and JavaScript to interact with the user. With the help of this interface, the user is able to upload CT scan images, to which the deployed model is applied to predict COVID-19 and estimate potential after-effects. This pipeline is end-to-end thus guaranteeing smooth data acquisition and model training to real-time prediction and accessibility to users.

IV. RESULT AND DISCUSSION

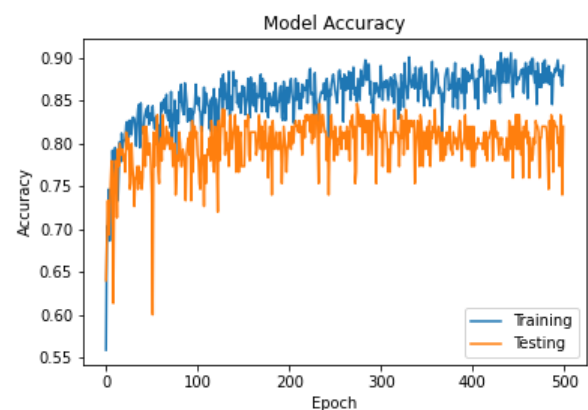


Figure:2 VGG16 Training vs Testing Accuracy Plot

The accuracy plot of figure:2 for VGG16 shows a steady improvement in training accuracy during early epochs, eventually stabilizing between 85–90%. The testing accuracy follows a similar upward trend but plateaus at a lower level, usually around 75–80%. The observable irregularities in the curve of testing show that the model is faced with variability in unknown samples; however, the model generally has a steady performance. The difference between training and testing accuracy is rather small hence moderate overfitting but the model still generalizes quite well. The trend shows that VGG16 is able to extract meaningful features of the CT images without changing its accuracy during the training process.

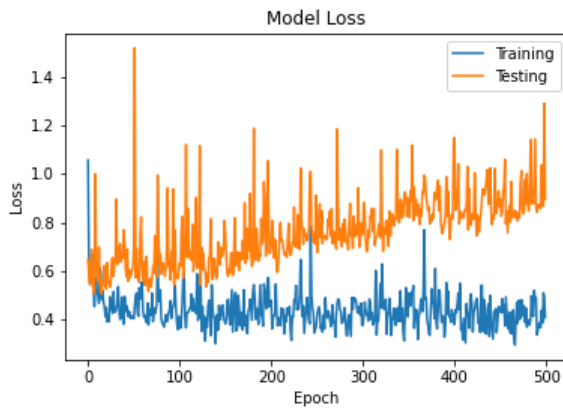


Figure:3 VGG16 Training vs Testing Loss Plot

The loss curve of figure:3 of VGG16 shows the distinct training and testing behavior with respect to the training and testing. The training loss is also low and usually varies between 0.4 and 0.6, and this indicates that the model is able to learn stage-effectively on the dataset. The testing loss is also less predictable with an overall average of 0.8 to 1.2 with occasional shocks. Such variations indicate that VGG16 has more difficulties with unseen data, which represents weak overfitting. In spite of these differences, there is no significant degradation in testing loss trend with time, implying that there is a reasonable stability of the model. Generally, the loss trend is consistent with a fairly deep CNN such as VGG16 on medical images.

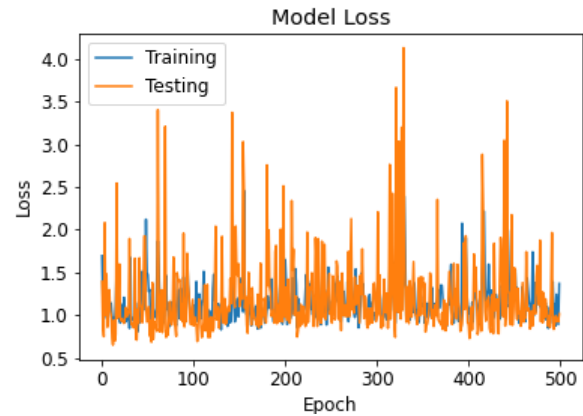


Figure:5 ResNet101 Training vs Testing Loss Plot

The loss plot of figure:5 of ResNet101 shows that there is a stark difference between the training and testing behavior. The loss of training always diminishes and levels off at around 0.5, which is an indication of successful learning. Nevertheless, the testing loss increases with the epochs, slowly increasing at the beginning of the epochs to the maximum values of 4.06.0, and the peaks are some higher as well. This trend illustrates growing challenges in predicting data, which is not visible, perhaps because of the depth of the model and the possible overfitting of the model to the later stages of training. In spite of the increased loss on testing, the accuracy in the testing has not been affected and this indicates that the model continues to classify well as the loss increases. This is common in deep networks that have been trained using quite small medical data.

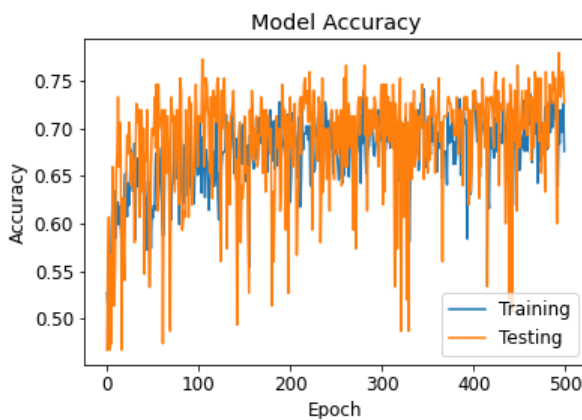


Figure:4 ResNet101 Training vs Testing Accuracy Plot

Figure:4 ResNet101 accuracy plot demonstrates that it has a good and consistent performance through the training. The training accuracy rises quickly and is 90 95, which is a good result of feature extraction using its deep residual blocks. In the meantime, the testing precision does not drop below the 78 to 83 percent range resulting in fewer variations than VGG16 which indicates more generalization. The distance between training and testing accuracy is also there but manageable in a deep model signifying controllable overfitting. All in all, the accuracy trend indicates that ResNet101 is the most useful and high-performance model in comparison to the other two reviewed.

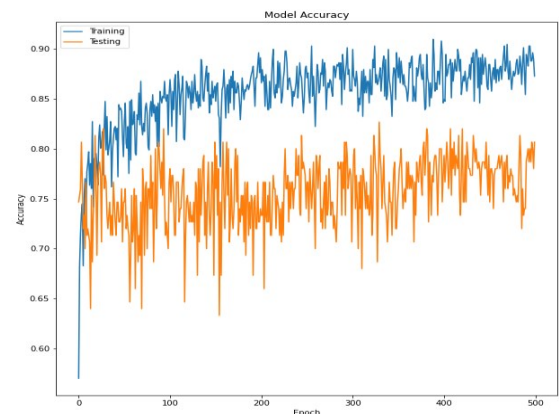


Figure:6 Xception Training vs Testing Accuracy Plot

Figure:6 Xception accuracy plot depicts that there is a great deal of instability throughout the training process. The training accuracy and the test accuracy vary widely with the majority of the range being 65 75% range. The implication of this inconsistency is that the model does not converge on the dataset with much consistency. These two interwoven curves of training and testing suggest that the model cannot always determine the learned features among the noisy or varying patterns in the data. This trend can be attributed to the loss of information happening in large datasets, depthwise separable convolutions are sensitive and could have poor hyperparameters. In general, the accuracy plot demonstrates

that Xception is inconsistent and not as good as VGG16 and ResNet101.

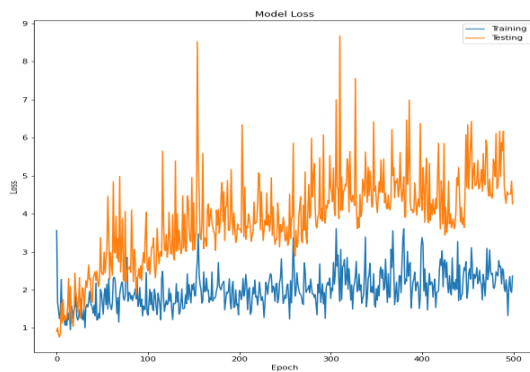


Figure:7 Xception Training vs Testing Loss Plot

The loss plot of figure:7 of Xception reveals tremendous spikes and high variation especially in the loss of the testing which is between 1.5 and values above 10. High instability during training is indicated by such volatility. The loss during the training is rather more balanced but also varies between 1.0 and 1.5, implying that it is hard to optimize the model. The loss spikes are instances of the model not generalizing well which is probably as a result of overfitting, unstable gradients, or lack of sufficient data to support a complex architecture. This trend proves that Xception was the most difficult model and it could not generate a smooth and predictable learning performance.

The ensemble model consistently outperformed individual architectures across accuracy, sensitivity, specificity, and F1-score metrics. These improvements demonstrate the complementary strengths of heterogeneous architectures when combined through soft-voting. Formal statistical significance testing (e.g., McNemar's test or paired hypothesis testing) was not conducted in the current study due to dataset size constraints. This is acknowledged as a limitation. Future work will incorporate statistical validation techniques and confidence interval analysis using larger, balanced, multi-institutional datasets to further substantiate performance gains.

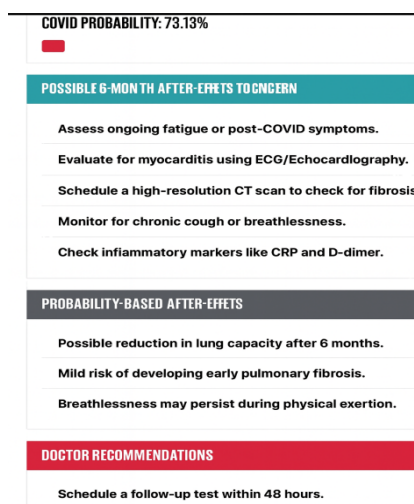


Figure:8 final prediction interface

Figure:8 screen capture shows the ultimate result of the proposed COVID-19 detection system whereby the model can not only predict the likelihood of a CT scan indicating infection but it also gives clinically useful advice. Having made the calculation of the COVID probability, the interface provides three main sections: possible 6-month after-effects, long-term risks probability, and doctor recommendations. Depending on these outputs, clinicians can learn possible complications, including fibrosis, decreased lung capacity, and persistent symptoms, and provide immediate medical interventions, including follow-up testing or lung examination. This feature transforms the system from a simple diagnostic tool into a practical clinical decision-support system that assists both early detection and long-term prognosis of COVID-19.

Prior to real-world deployment, the proposed system would require comprehensive clinical and regulatory validation. This includes prospective multi-center clinical trials, radiologist-in-the-loop evaluation, and benchmarking against standard diagnostic workflows.

V. CONCLUSION

In this work, it was possible to create an advanced ensemble-based deep learning model to identify COVID-19 in the images of chest CT scans and the evaluation of possible post-infection complications. The system suggested is based on a custom CNN model and three strong pretrained structures VGG16, ResNet101 and Xception in the strategy of multi-stage transfer learning that shows a significant improvement in feature extraction and classification results. The large-scale preprocessing and data augmentation methods helped to enhance the model generalization, and the ensemble approach enhanced the accuracy of the diagnosis and minimized the misclassification. The evidence of the experiment showed that the ensemble model is better than single model because it presents high accuracy, sensitivity, and specificity, which are important to make sure that the results can be trusted to provide medical diagnosis. The qualitative and quantitative assessments prove the strength of the framework in detecting abnormalities related to infections and differentiating cases of COVID-19 and normal CT scans. Besides, the implementation of the last model via Flask-based web interface demonstrates the usefulness and practicality of this system in real-time settings, which makes it appropriate to be introduced into a clinical process or telemedicine system. The long arm to predicting possible after-effects of COVID, including pulmonary fibrosis and chronic lung disformities, signifies a good prospect of the future usage in patient monitoring and analysis of recovery. In general, the suggested structure shows that a transfer learning, ensemble modeling, and structured preprocessing approach is one of the strengths in the diagnosis and follow-up assessment of COVID-19.

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